MICROSOFT FABRIC
USER GROUP AUSTRIA

MICROSOFT FABRIC
& POWER BI
MEETUP #6
GRAZ EDITION
19. März. 2026

Mastering Power BI Model Types

Import-, DirectQuery- or Composite models?

Who am I

Or who makes that noise…

- Gregor Prohaska
- Operations Manager @ COSMO CONSULT
- „Some“ years of experience

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- Father of 2
- Curious by nature
- Don’t take myself to seriously ;-)

AGENDA

01 Terms and Concept

02 Import Model

03 DirectQuery Model

04 Direct Lake Model

05 Composite Model – Import + Direct Query

06 Composite Model – Direct Query for Power BI semantic models

07 Composite Model – Import + Direct Lake on OneLake

08 Which one to use…

Terms and Concepts

Relationships

Semantic Models

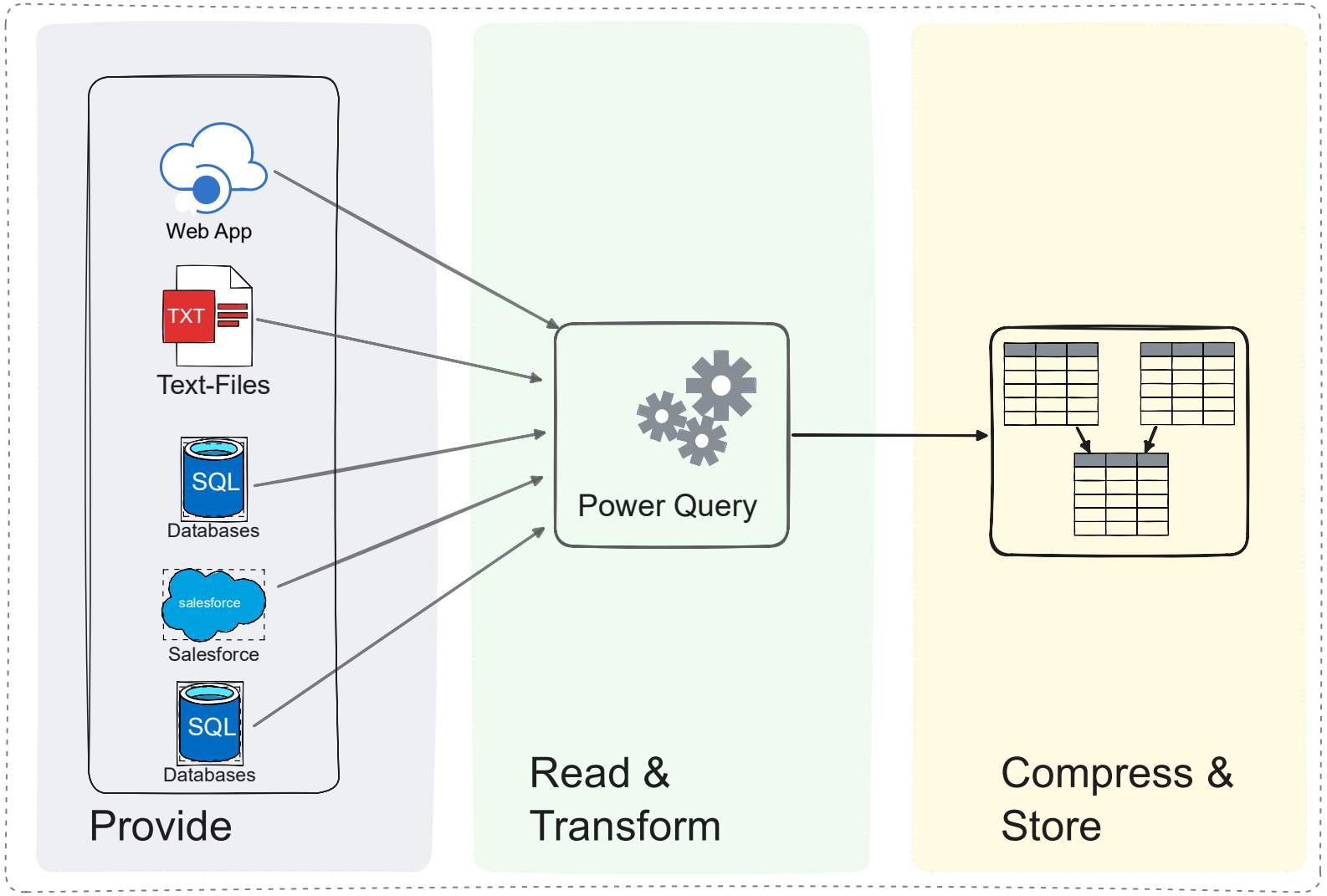
Source Group

Storage Modes

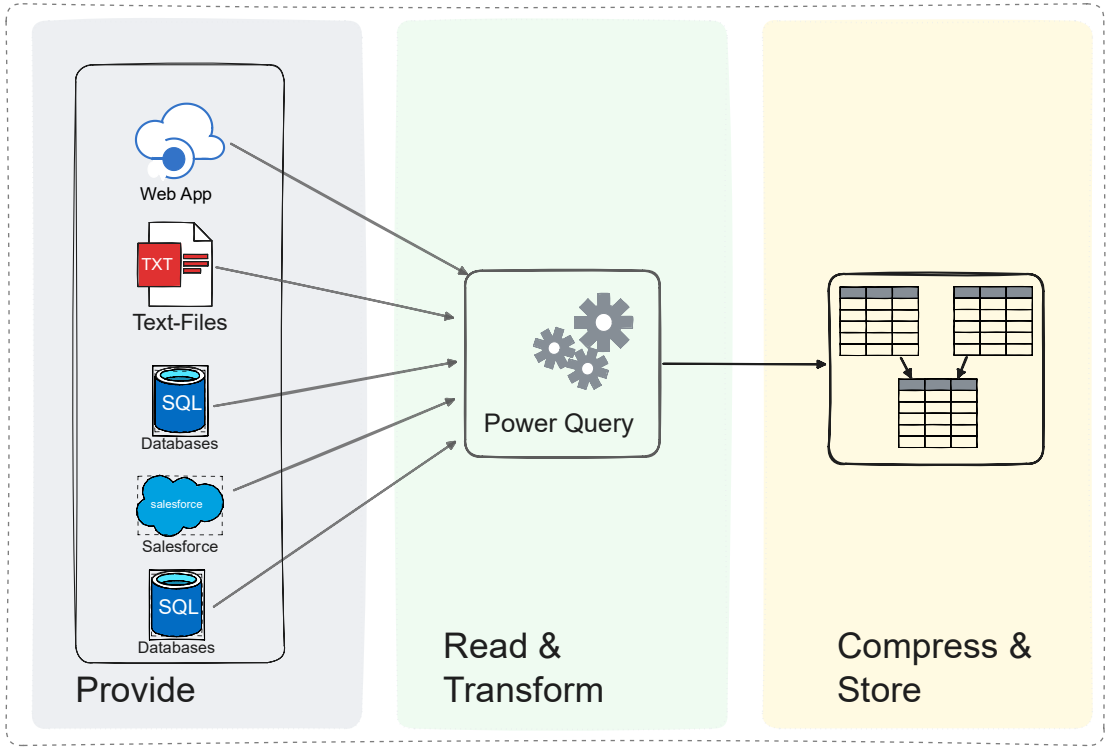
Security

Performance

Import Model

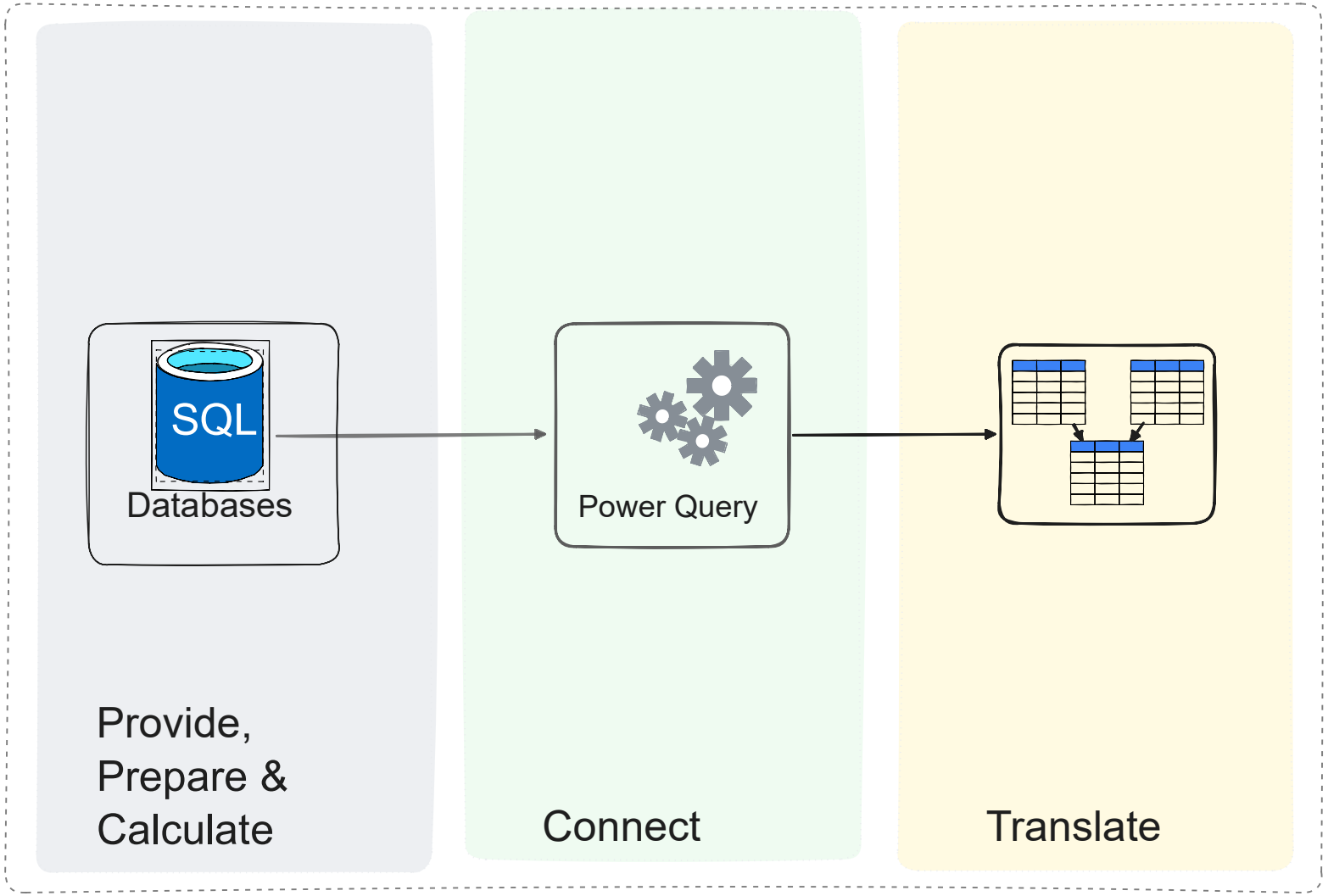


Import Model

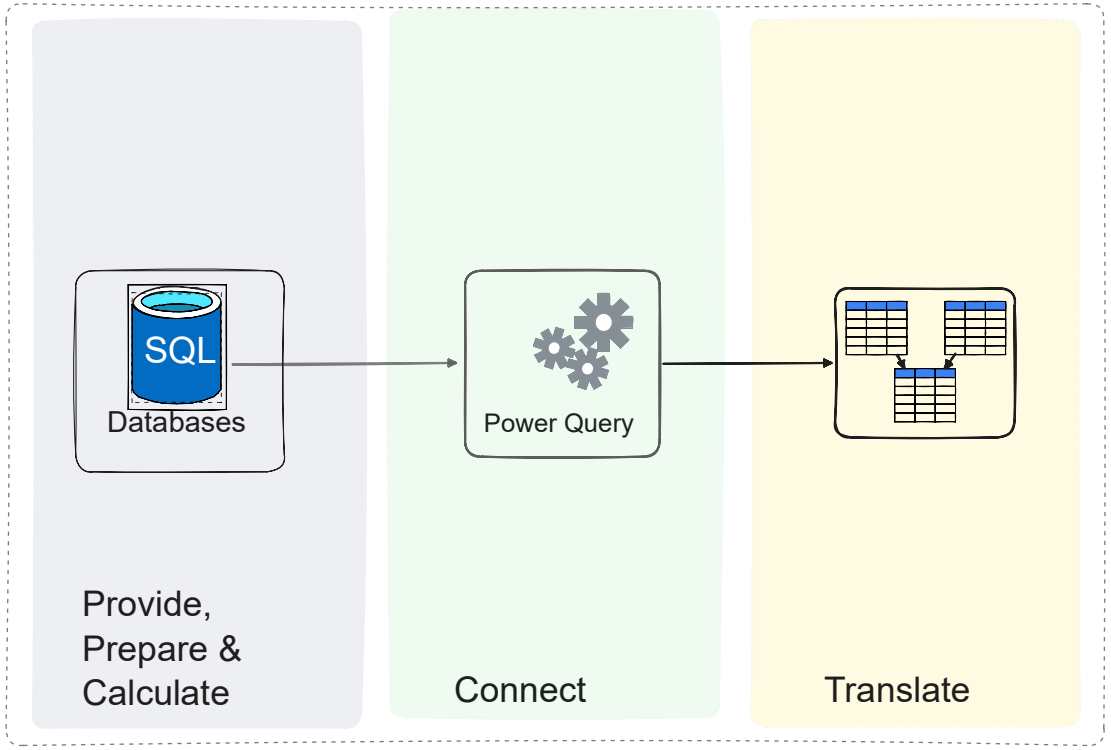


- Supported by all connectors/data sources
- Vertipaq engine
- Limited storage & (scheduled) refreshes
- Decouples import and query workload
- Semantic model security only
- (mostly) regular relationships

DirectQuery Model

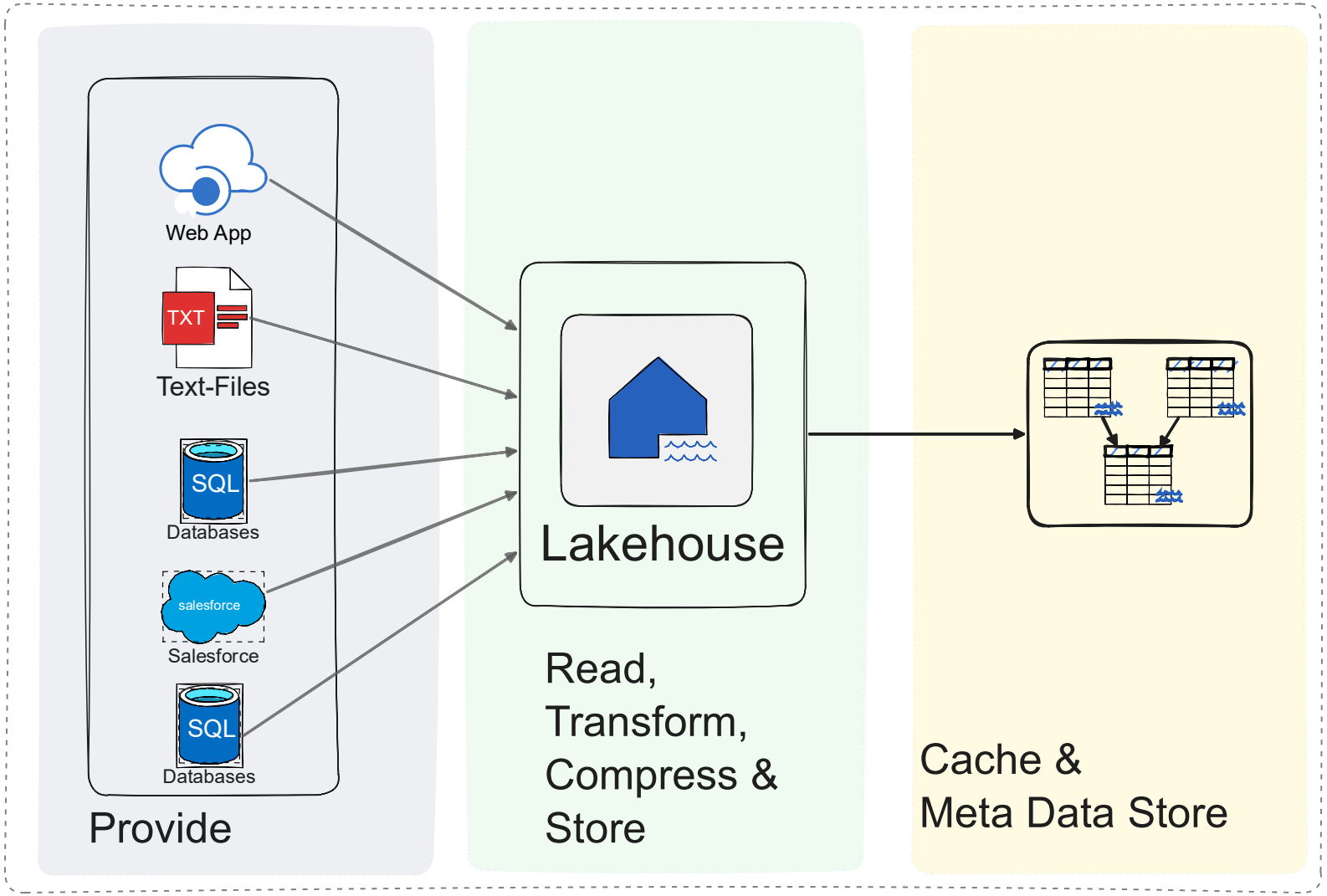


DirectQuery Model

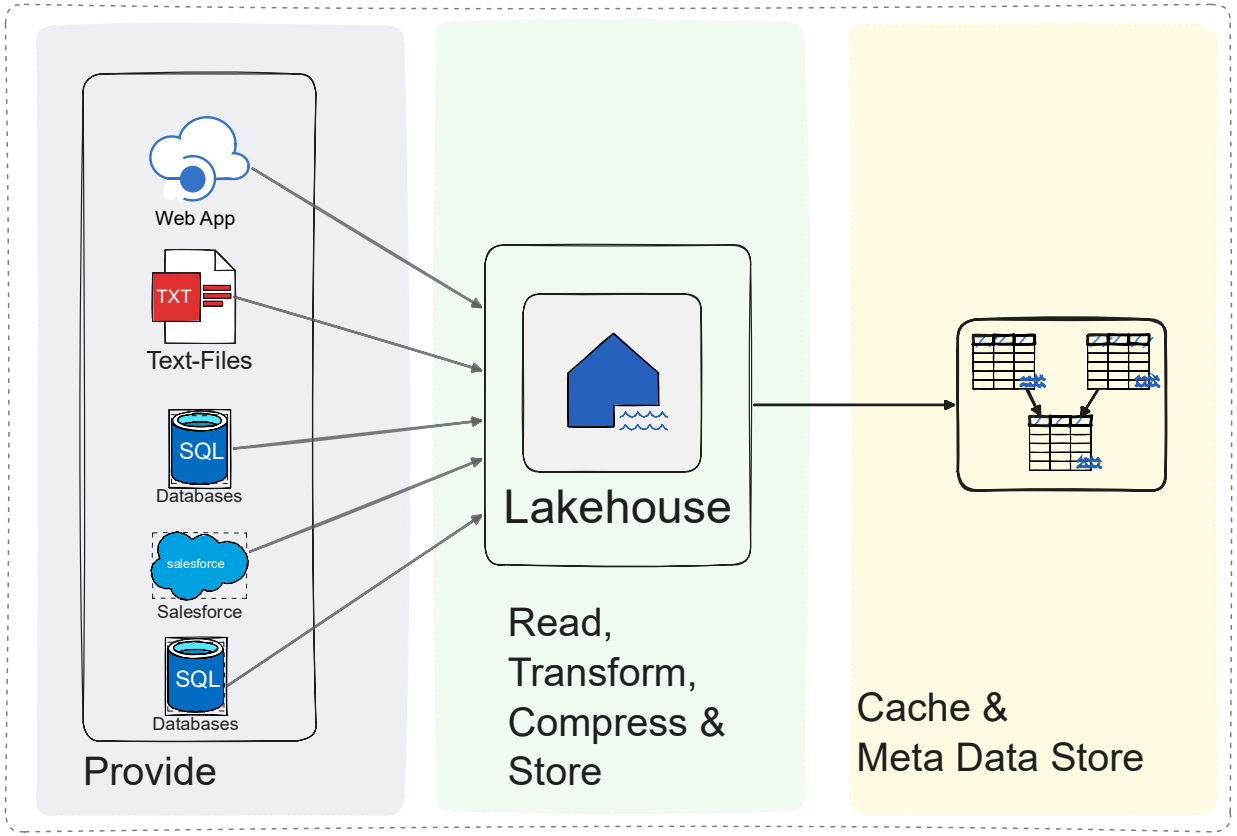


- Data stays at source & always up-to-date
- „no“ model size limits
- Data source security can be leveraged
- Limited DAX & data source support
- Calculated tables not supported

Direct Lake Model



Direct Lake Model



- DAX fully supported
- Vertipaq engine
- Decouples import and query workload
- Model size limited by fabric capacity
- Data source security can be leveraged
- Limited support on calculated table

Composite models

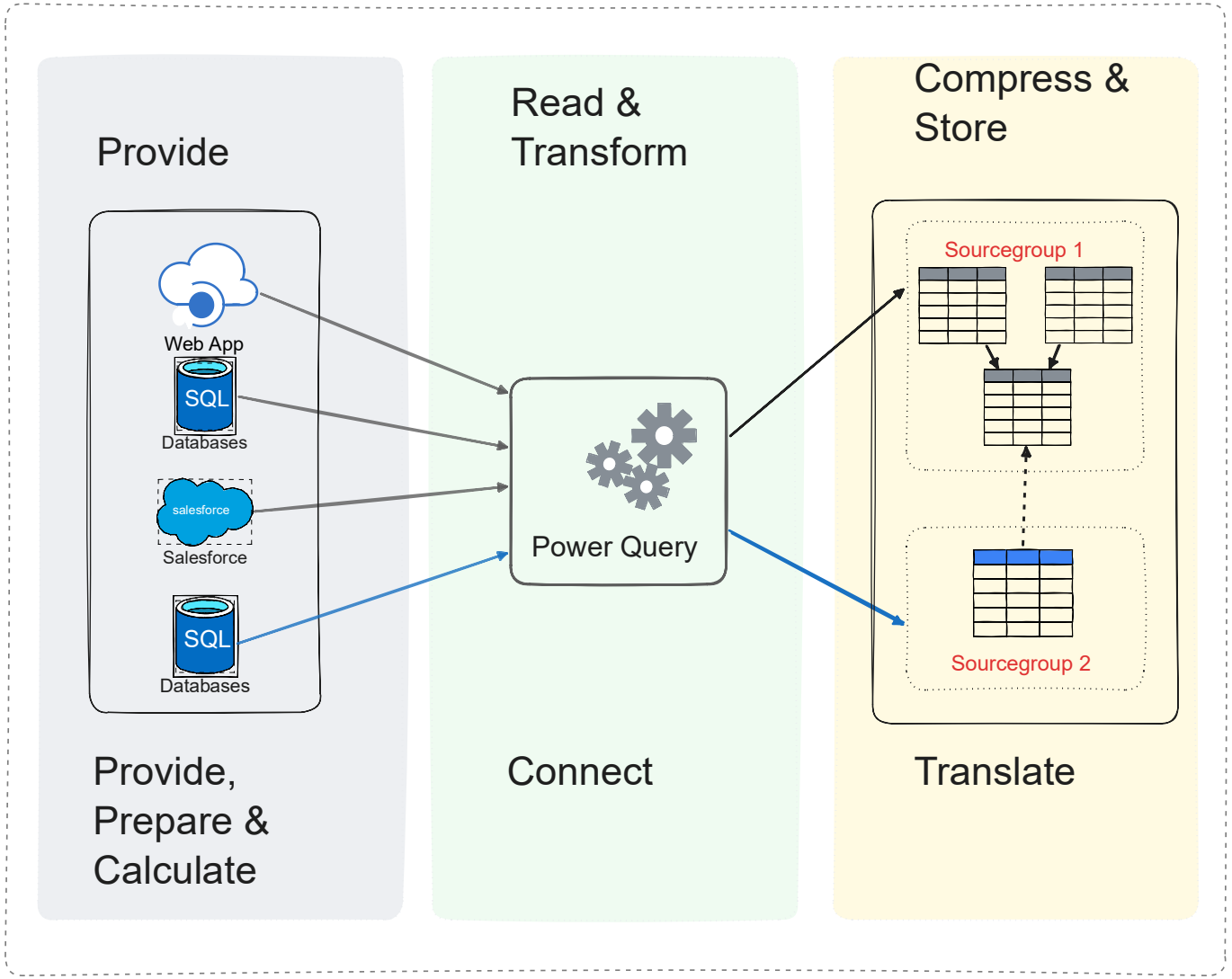
One-Model

- DirectQuery tables comming from different data sources
- Import and DirectQuery tables in the same semantic model
- Import and Direct Lake on OneLake tables in the same semantic model
- DirectQuery and Direct Lake on OneLake tables in same semantic model

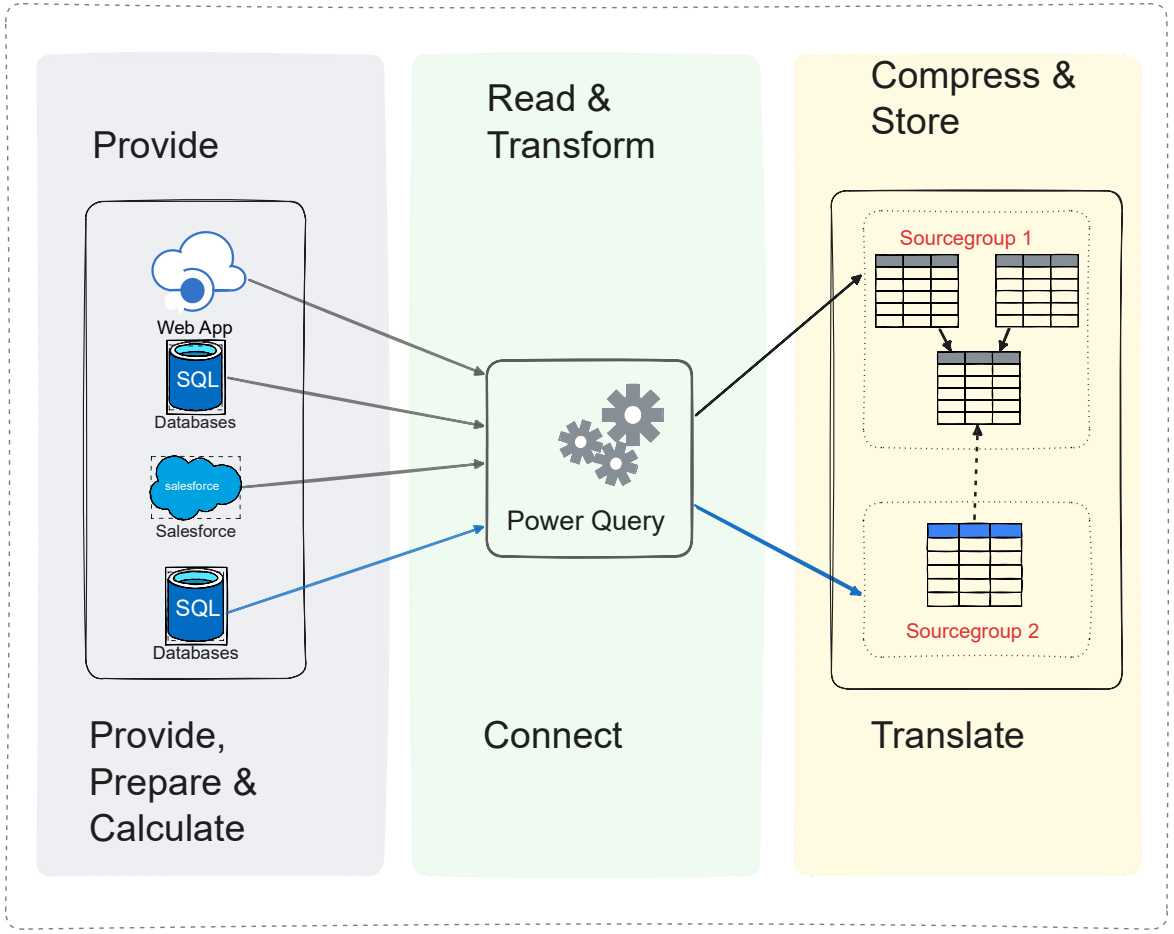
Multi-Model

- Combining multiple semantic models
- Direct query to another semantic model with additional tables in import or DirectQuery storage modes

Composite Model – Import + DirectQuery



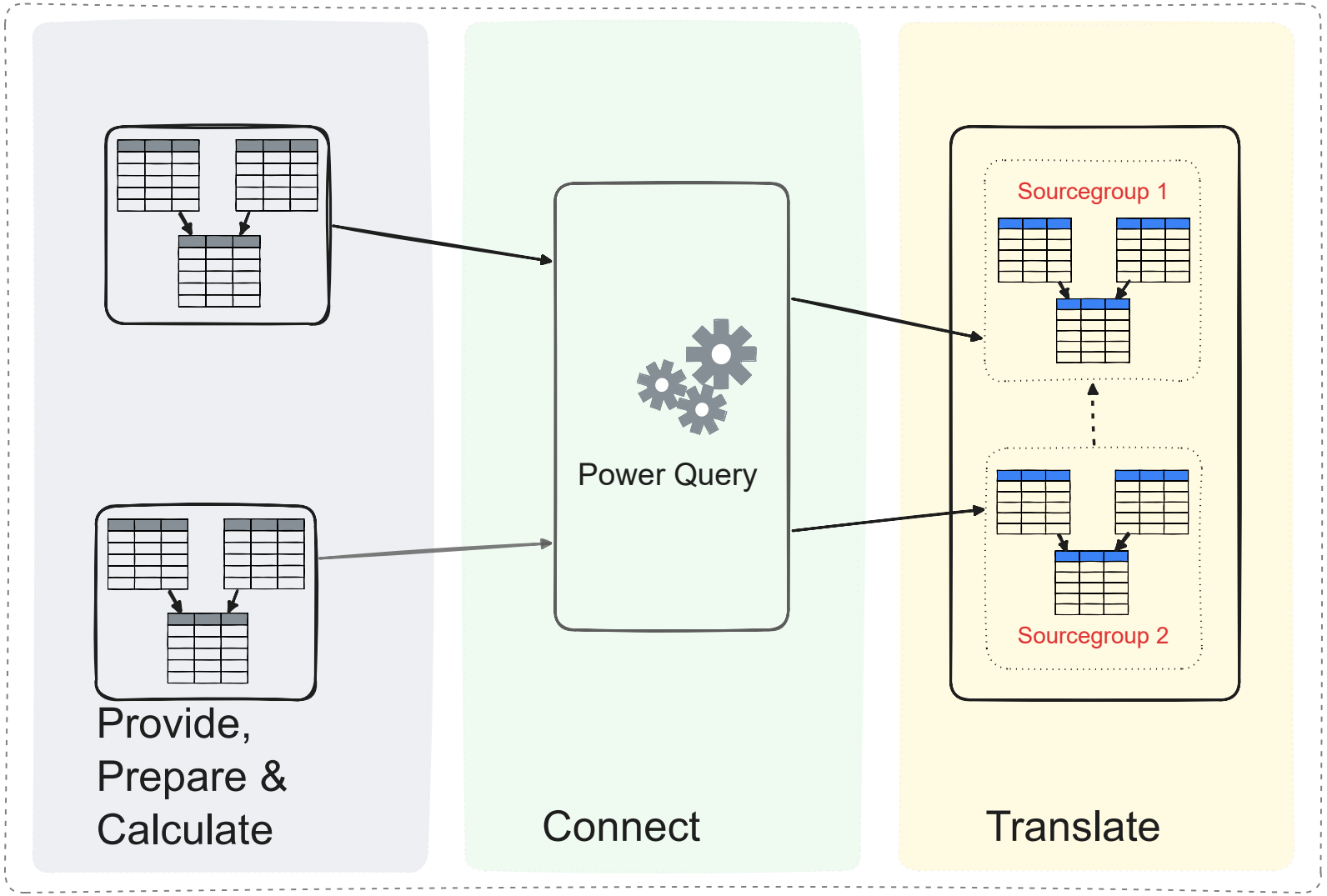
Composite Model – Import + DirectQuery



- Import and DirectQuery-Tables
- Dual-Mode possible (regular relationships)
- Limitations of Direct Query-models applies for DQ-Part
- Data source security can be leveraged for DQ-tables

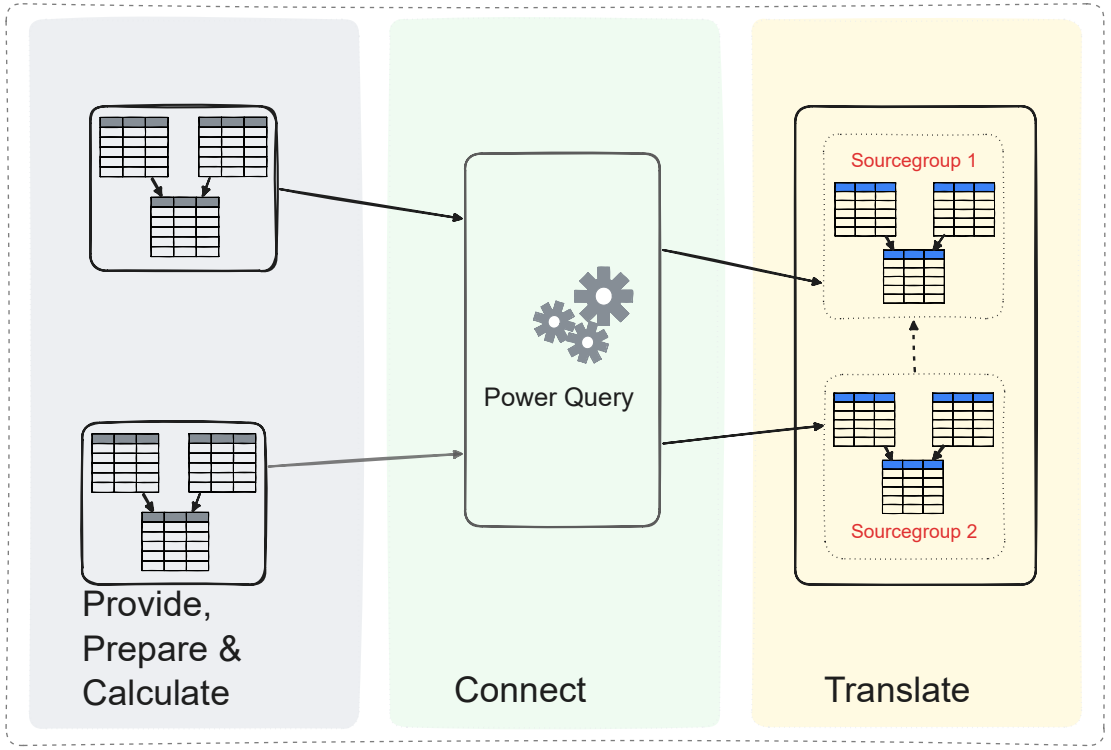
Composite Model – Direct Query for Power BI semantic models

… and Analysis Services 2022+



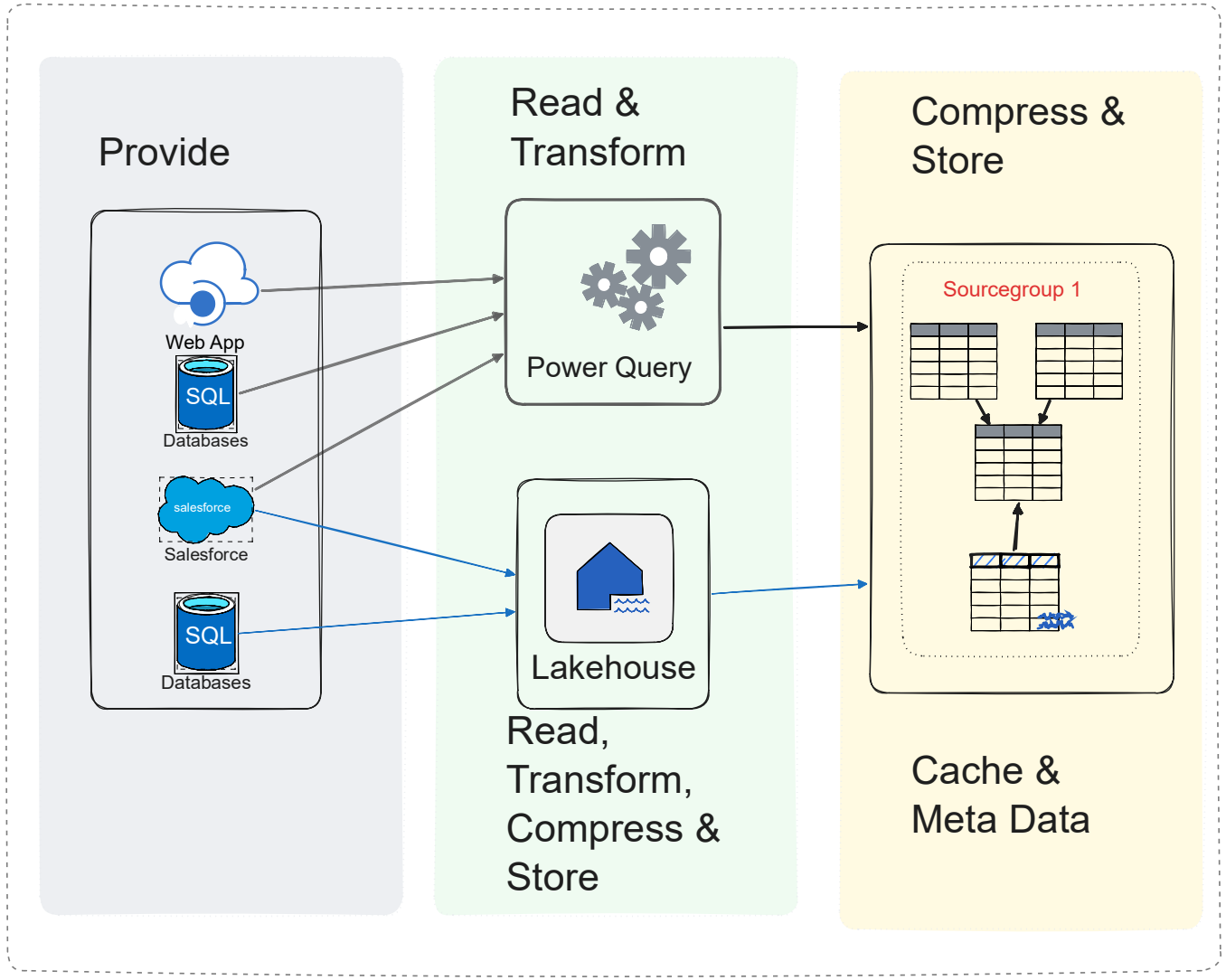
Composite Model – Direct Query for Power BI semantic models

… and Analysis Services 2022+

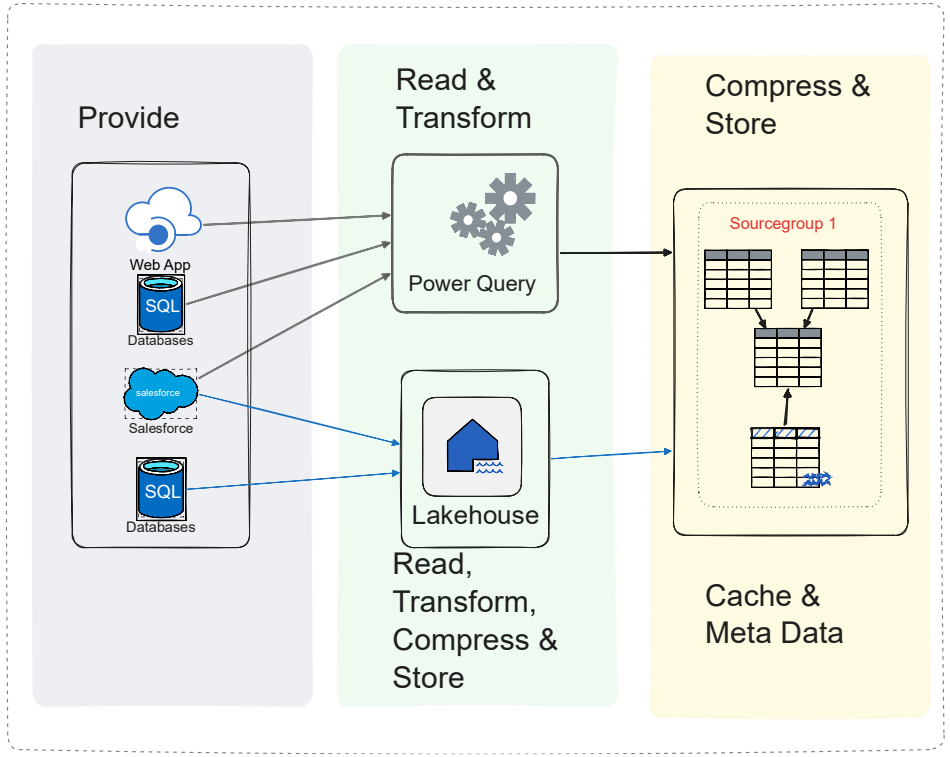


- Source Models can be of any type
- Multiple semantic models can be combined in one
- Further Data can be added (Import and DQ)
- Source semantic model security is applied
- Consumers need READ-Permisson on all models
- Chaining possible (max. 3 Levels)

Composite Model – Import + Direct Lake on OneLake



Composite Model – Import + Direct Lake on OneLake



- Contains Import and Direct Lake Tables
- One Sourcegroup only
- Regular relationships
- Significant improvement over „Multi-model models“
- Fabric only
- All limitations of Direct Lake apply to DL-Tables

Usage Scenarios

	Importmode	DirectQuery	Direct Lake	Composite – IM + DQ	Composite – IM + DL	Composite – DQ on PBI\AS
Best used when	Most flexible solution; use it for „basic" models or if you have to cover diverse data sources. Best in query performance at the costs of data freshnes	Best for data freshnes on relational or multi-dimensional data sources at the costs of query performance	Best for data freshnes if data resides in OneLake already. Use it for irregular & seldomly used models in Fabric	Use it in planning/ write-back scenarios or to build aggrega-tions to overcome size limitations	Best on huge fact tables with dimensions which have to be adjusted in the model (e.g. calculated columns)	Use it for cross model reporting or to enable self-service enrichment

Thank you.

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stay curious &
please give feedback

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Slides: https://bit.ly/PowerBIModeltypes20260319